

Topic C7

Acids, bases and salts

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

Syllabus map

c7.1	Characteristic properties of acids and bases
c7.2	Acidity, alkalinity and indicators
c7.3	Preparation of salts
c7.4	Ionic equations and titration

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. Acids, bases and alkalis

Acids produce $\text{H}^+(\text{aq})$ ions; bases neutralise acids; alkalis are soluble bases that produce $\text{OH}^-(\text{aq})$. Acid + metal gives salt + hydrogen; acid + base gives salt + water; acid + carbonate gives salt + water + carbon dioxide. The salt name comes from the acid: hydrochloric forms chlorides, nitric forms nitrates, sulfuric forms sulfates.

Examiner insight

Choose reactants from the required salt: the acid supplies the anion and the metal/base supplies the cation.

2. pH and indicators

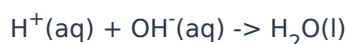
The pH scale describes acidity and alkalinity. At 25 °C, pH below 7 is acidic, 7 neutral and above 7 alkaline. Universal indicator estimates pH across a colour range; litmus distinguishes acid from alkali; methyl orange and phenolphthalein give sharper end-points in titration. A lower pH means a higher H^+ concentration.

Examiner insight

Strength describes ionisation; concentration describes amount per volume. A dilute strong acid is still fully ionised.

3. Neutralisation and ionic equations

Neutralisation removes acidic H^+ ions with basic OH^- ions to form water. Spectator ions remain unchanged and are removed from an ionic equation. Metal oxides, hydroxides and carbonates neutralise acids.



Worked example

Sulfuric acid + copper(II) oxide $\rightarrow$ copper(II) sulfate + water.

Examiner insight

When writing ionic equations, balance both atoms and total charge.

4. Making soluble salts

For a soluble salt of an unreactive metal, warm acid and add excess insoluble base/carbonate until no more reacts. Filter off excess solid, evaporate some water, cool to crystallise, filter and dry. For sodium, potassium or ammonium salts, use titration because all their common bases and salts are soluble.

Examiner insight

Do not evaporate to dryness: crystals form better from a hot saturated solution on cooling.

5. Making insoluble salts and titration

An insoluble salt is made by precipitation: mix two suitable soluble salt solutions, filter the precipitate, wash with distilled water and dry. In titration, use a pipette for a fixed volume, burette for variable volume and indicator for end-point; repeat without indicator to obtain pure crystals.

Examiner insight

A conical flask may be rinsed with distilled water because added water does not change the moles being titrated.

Exam technique

High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

Practice Paper

Topic C7

cids, bases and salts**Instructions**

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which ion causes aqueous acidity? [1]
- A H^+
 - B OH^-
 - C Na^+
 - D Cl^-
2. Acid + carbonate produces salt, water and [1]
- A hydrogen
 - B oxygen
 - C carbon dioxide
 - D chlorine
3. Sulfuric acid forms [1]
- A chlorides
 - B nitrates
 - C sulfates
 - D carbonates
4. Which pH is strongly alkaline? [1]
- A 1
 - B 5
 - C 7
 - D 13
5. What is the ionic equation for neutralisation? [1]
- A $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$
 - B $\text{Na}^+ + \text{Cl}^- \rightarrow \text{NaCl}$
 - C $2\text{H}^+ \rightarrow \text{H}_2$
 - D $\text{O}_2 + 4\text{e}^- \rightarrow 2\text{O}^{2-}$
6. Which apparatus delivers a variable accurate volume? [1]
- A beaker
 - B burette
 - C measuring cylinder
 - D evaporating basin
7. Why is excess copper(II) oxide added to acid? [1]
- A to use all the acid
 - B to make gas
 - C to change the anion
 - D to form sodium ions

8. How is excess insoluble solid removed? [1]
- A distillation
 - B filtration
 - C chromatography
 - D titration
9. How is barium sulfate prepared? [1]
- A crystallisation only
 - B precipitation from soluble salts
 - C electrolysis of molten salt
 - D heating barium metal
10. Which salt normally requires titration for preparation? [1]
- A copper sulfate
 - B magnesium chloride
 - C sodium nitrate
 - D zinc sulfate

Section B · Structured questions

11 Dilute hydrochloric acid reacts with magnesium.

(a) State two observations. [2]

(b) Name the salt formed. [1]

(c) Write the symbol equation. [2]

(d) Describe the test for the gas and positive result. [3]

[Total: 8]

12 Copper(II) sulfate crystals are prepared from sulfuric acid and copper(II) oxide.

(a) Explain why copper metal is unsuitable. [2]

(b) Describe addition of copper(II) oxide. [2]

(c) Explain the filtration step. [2]

(d) Describe crystallisation and drying. [2]

[Total: 8]

13 25.0 cm³ sodium hydroxide is titrated with hydrochloric acid.

(a) Name the apparatus for the sodium hydroxide. [1]

(b) Name the apparatus for hydrochloric acid. [1]

(c) Describe how a reliable mean titre is obtained. [3]

(d) Explain how pure sodium chloride crystals are then made. [3]

[Total: 8]

14 Lead(II) sulfate is an insoluble salt.

(a) Name two soluble solutions that can produce it. [2]

(b) Write the ionic equation. [2]

(c) Describe how to obtain a pure dry sample. [3]

(d) State the name of this preparation method. [1]

[Total: 8]

15 Compare a strong acid with a weak acid of the same concentration.

(a) Define strong acid. [2]

(b) State which has lower pH. [1]

(c) Predict which reacts faster initially with magnesium. [2]

(d) Explain why both may produce the same final hydrogen volume. [3]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

Solutions · Multiple choice

- 1 A. Acids release hydrogen ions in water.
- 2 C. Carbonates release carbon dioxide with acids.
- 3 C. The acid anion becomes sulfate in the salt.
- 4 D. Values well above 7 are alkaline.
- 5 A. Hydrogen and hydroxide ions form water.
- 6 B. A burette records accurately delivered volume.
- 7 A. Excess insoluble base ensures neutralisation.
- 8 B. The solid remains as residue.
- 9 B. It is insoluble and precipitates when suitable ions meet.
- 10 C. Sodium compounds and the alkali used are soluble.

Solutions · Structured questions

11. 11(a) Effervescence and magnesium disappears / temperature rises [2]. (b) Magnesium chloride [1]. (c) $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$ [2]. (d) Apply lighted splint [1]; squeaky pop [1], showing hydrogen [1].
12. 12(a) Copper is below hydrogen / does not react with dilute sulfuric acid [2]. (b) Warm acid; add solid until excess remains [2]. (c) Remove unreacted excess oxide as residue [2]. (d) Evaporate to saturation, cool, filter and dry crystals [2].
13. 13(a) Pipette [1]. (b) Burette [1]. (c) Rough titration then repeats to obtain concordant titres and mean [3]. (d) Repeat measured volumes without indicator, evaporate, cool, filter/dry [3].
14. 14(a) Lead(II) nitrate and sodium/potassium sulfate [2]. (b) $\text{Pb}^{2+} + \text{SO}_4^{2-} \rightarrow \text{PbSO}_4$ [2]. (c) Mix, filter, wash precipitate with distilled water, dry [3]. (d) Precipitation [1].
15. 15(a) Completely ionised in aqueous solution [2]. (b) Strong acid [1]. (c) Strong acid due to higher H^+ concentration [2]. (d) Same acid moles / both eventually supply same H^+ if metal excess, so same moles H_2 [3].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.